6CS014

Portfolio Task 2

AI Driven Autonomous Delivery Drone with Dynamic Pathfinding

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ABSTRACT

Autonomous drones are a concept old before time but their proper implementation has yet to see the world. This report presents the ideas of various implementation and designs of autonomous drones used throughout the world for delivering capabilities. Most of the systems utilize artificial intelligence to navigate their way and avoid obstacles along the way. They dynamically find their way and adapt accordingly to whatever the adversary maybe. By integrating GPS sensors, logistic data, terrain information and pathfinding algorithm, the drone can efficiently and safely travel and delivery the package. Some drones implement a different approach where it finds the desired path and beelines straight to the goal. This report compares all different techniques that can be implemented while discussing their pros and cons with real world data.

1. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) are the bleeding edge technology that are constantly revolutionizing the last-mile logistics. Their exceptional flexibility allows them to bypass constraints like road traffic and provide point-to-point delivery efficiently. The anatomy of a UAV commonly implements AI-driven path finding advanced algorithms that can dynamically adapt to the different hurdles the agent may encounter along its way while respecting the primary objective of delivering the package safely with energy constraints. Recent surveys emphasize on machine learning and neural networks that can progressively adapt to environments in real time. These innovations have proven to be pivotal when navigating dense urban areas where proper coordination and avoidance is required for a reliable delivery. (Meng, 2025)

As industrialization takes a hold of the society, the urgency of improving drone delivery has grown due to congestion and demand for rapid service. The conventional way of delivering using ground vehicles leads to even more congestion and excessive carbon emission. Whereas, AI-powered UAVs can dynamically adapt the different routes and the constantly changing weather. In fields like medicine, time is of the essence and it is imperative the medical supplies reach the destination in time and with safety. Given these constraints, the AI-powered drones that can work around the clock are a perfect fit for these high-stake tasks. Given the constant demand, the development of these agents is a pressing research area and one that promises to reshape the urban last-mile in logistics. (Meng, 2025)

2. AIM & OBJECTIVE

- **AIM:** To research autonomous delivery drone that inherits an AI-driven framework with the ability to dynamically determine an optimized route with the goal of improving efficiency, reliability and sustainability in last-mile logistics.

- **OBJECTIVES:**

    1. Investigate the current state of autonomous drones and their applicability in last-mile delivery.

    2. Compare different methods used in the agents that aims to increase reliability, efficiency and minimize emission.

    3. Research different route planning techniques like genetic algorithms and reinforcement learnings to improve drone coverage and responsiveness.

    4. Assess the integration of AI-powered drones with real world constraints like weather conditions, urban infrastructure and obstacle avoidance.

3. LITERATURE REVIEW

3.1 Trajectory Optimization with Temporal Logic

(Chen, 2025) proposes a trajectory method for UAV based medical drones under strict time windows. The author lays out a scenario where a drone must deliver blood packages to several hospitals where each hospital has a different order of priority and time window. These delivery tasks are encoded into Signal Temporal Logic (STL) allowing specification of these constraints. Buildings and objects are modelled as 3D convex shapes that the agent must avoid at all cost. This is ensured using the Convex Feasible Set (CFS). The entirety of combining UAVs with STL goals and obstacle avoidance is casted as the convex optimization problem. Using this method, a safe and dynamically feasible flight paths can be computed that satisfies the given constraints. When this solver is tested in a simulation, it can reliably compute an optimal path that respects the given deadlines while avoiding obstacles. This paper's framework guarantees that the drone satisfies both the primary goal as well the constraints in a time-critical medical logistics. However, the fact remains that all the evidence provided are in a hypothetical environment and the paper lacks real-world evidence hence requiring further investigation to fully evaluate its scalability, reliability and robustness when faced with real-world uncertainties.

3.2 Multi-Drone Energy-Aware Delivery

(Qin, 2025) lays out an interesting conundrum where multiple drones are arranged in a time sensitive setting. The catch is that each drone has a limited battery life and they must make the most of it and achieve the primary goal. Each drone must carry multiple packages and plan a route that is energy efficient and minimizes consumption. The author tackles this problem by dividing it into various sub-tasks. First, he uses K-means to partition the different area of the environment into different regions. Then reinforcement learning is applied to determine the optimal path payload for each drone. Finally, they conclude by proposing an actor-critic based multiple-agent RL algorithm that coordinates the drones. This method also referred to as Optimized Plan Selection (MAR-OPS) integrates clustering, reinforced learning and optimization all into a single framework. Extensive real-world testing shows that Qin's approach outshines every other framework that primarily aims to reduce consumption. This learning-based framework can also adapt to the consumer need and handle different payloads accordingly without sacrificing optimization. Nevertheless, while the results maybe impressive, the sheer complex nature of the framework and the raw computation power required might prove to be a challenge where the environment and drones are highly constrained. Further improvements could be made so that agents that operate on a simpler rubric could benefit from this framework while still being able to leverage the efficiency and robustness of the framework.

3.3 Hybrid GA & Bayesian Networks for UAV Logistics

(Mahmoodi, 2025) introduces a multi-objective model for drones under uncertainties. The model works like every other standard point-to-point delivery model with the core difference of it being able to handle multiple objectives. The multiple objectives could either be different delivery locations or a restock station where the drone would refuel itself. Unlike other models, this model explicitly incorporates uncertain traffic conditions. It then uses Bayesian Belief Networks to assess the risk of delay for each route segment. This allows the model to make proactive choices in real-time and allows it to re-route itself from the primary objective and take a detour to a secondary objective. This ensures that the agent is very reliable and never gets lost in the transit. This capability does trade off the efficiency of the agent where the agent has to constantly reassess the route based on the given constraint. Another case where this can be sub-optimal is in situations where the model overestimates the risk. To mitigate this, a hybrid approach can be used applying the NSGA-II genetic algorithm. This ensures a very reliable model that will always reach the destination while taking variable paths along the way that may or may not be sub-optimal. The author also demonstrates simply using a standard genetic algorithm without any Bayesian networks. The comparison shows that the hybrid model incorporating the Bayesian networks have a much lower average risk thus balancing reliability and robustness.

3.4 Vehicle UAV Integrated Routing

(Ghaffar, 2024) explores a scenario where a truck and a drone both are used as the delivery agent. The report proposes to utilize both agents to access hard to reach areas leveraging their full capabilities. This dual-mode framework is first clustered by DBSCAN. They are divided as “drone only” and “vehicle accessible” location. The ground vehicle is given the priority and the drone is sent off to handle remote clusters where the ground vehicle would prove to be worthless. For the route optimization, a Genetic Algorithm and Artificial Bee Colony (ABC) algorithm are compared and applied. The Artificial Bee Colony algorithm is generally proven to be faster but is limited by its tendency to be stuck in a local-optima. This severely hinders its performance so to combat this; it is augmented with Simulated Annealing. The resulting ABC-SA algorithm yields a better coverage and results a better-quality solution. Simulated scenarios show that the enhanced algorithm reduces the overall computation time and improves satisfaction rate of the constraints over the standard methods. Overall, studies like these often prove that combining traditional autonomous delivery agents with hybrid optimization of pre-existing algorithms result in a significantly better system with enhanced robustness, scalability in dynamically changing complex environments.

3.5 Edge AI for Real-Time Path Planning

(Pasupuleti, 2025) investigates the uses of on-board Edge AI as a mode of data transmission to the agents as compared as the contemporary solution to the tradition cloud. The traditional cloud-based navigation heavily suffers from low latency. Since the agents themselves are primarily focused on route optimization, they often rely on the cloud to make primary decisions and get additional information like the weather forecast, terrain location, incoming traffics, delivery destinations and so on. This architecture has been in practice since the dawn of time due to its simplicity and reliability. But the author proposes a bold change of deploying a hybrid edge-cloud architecture. In this architecture, an NVDIA Jetson Nano GPU is embedded on the drone itself allowing it to locally make decisions on the fly. This allows the drone to proactively make quick decisions with much lower latency as everything is computed by itself while it still relies on the cloud for additional data and global updates. In simulations, the drones were able to navigate dense urban areas and avoid obstacles with a much greater accuracy using the on-board RL model. The results are very compelling as the re-routing time and drastically decreased. Furthermore, regression analysis proves that in the long run the efficiency is also improved due to the overall decreased latency. However, since the computation is happening internally, this can severely impact the battery life of the agent. Reports show the battery life to be cut down by 300% depending on the environment and the amount of re-routing required.

3.6 Multi-Platform Routing with En-Route Charging

(Anon., 2025) introduces a multi-platform routing problem that integrates the use of trucks, drones and delivery robots. In this case, the truck is primarily used as a mobile base of operations. It is used for charging and launching drones and robots en route. This model also covers other scenarios where the same drone or robot can be used to deliver to multiple customers in a single trip saving fuel and energy and reducing the overall emission. The UAVs are also designed to be flexible enough to dock to any truck at any time. Crucially, while in the truck, UAVs can enter recovery mode to further conserve their battery. The author coins this Vehicle-Robot-Drone Problem (VRP-DR) as a program that minimizes operation cost. Numerical experiments indicate that this hybrid system yields substantial time and cost saving as compare to a single platform routing agent. Further insights are also provided to further improve the efficiency and throughout of the system like increasing the fleet size of the truck so that it can fit a greater number of UAVs though this would require a very large truck capable of hoarding resources for all of the agents. This paper's novelty is combining different autonomous agents and integrating them into a single system for an efficient and cost-effective delivery operation for last-mile networks that would otherwise not be possible with single platform agents.

3.7 Systematic Review of Drone Delivery

(Jazairy, 2025) provides a board overview of the landscape of autonomous drones in last-mile networks and the endeavors taken to enhance its future scope. After a systematic review of over 300 publications, the author synthesizes its potentials, challenges and solutions of drone delivery from a logistic standpoint. One of the key-finding is that drones are able to greatly minimize the emission under ideal conditions but faces limitations like limited batter reservoir, payload weight limit and regulatory hurdles. The review highlights potential solutions like sustainable battery, smart re-routing, energy-efficient-routing, multi-modal integrations and so on. Furthermore, it identifies the gaps in research and in practice especially in safety protocols of the UAV itself as well the pedestrians it may encounter. The possibility of integration of edge AI cloud technology and coordinated fleet system is also highlighted. This comprehensive overview offers a holistic roadmap of UAVs and motivates further study in this area to realize the full promise of a near-perfect autonomous delivery system. Additionally, the study also emphasizes the importance of collaboration between the policymakers and developers to accelerate the real-world adoption.

Maintaining a high public acceptance and addressing all the ethical concerns is deemed equally essential for the fruition of such system.

4. ANALYSIS OF FINDINGS

- (Chen, 2025): Trajectory optimization of drones with STL encoding produces a provable and safe routes in a time sensitive environment for medical UAVs. Their convex programming for collision avoidance combined with priority driven objectives ensures all the goals are met within the given constraints.

- (Qin, 2025): A multi-agent deep RL framework yields an energy efficient multi-drone routes that minimizes delivery delays and conserves energy. By coordination the multiple agents, the framework clusters regions and assigns routes and payload as per the agent's constraints ensuring a robust and reliable last-mile network.

- (Mahmoodi, 2025): The hybrid model of NSGA-II with Bayesian models balances completion of the objective under real-world uncertainties like traffic and weather conditions. Incorporating the Bayesian network does ensure the completion of the objective albeit at the cost of increased latency.

- (Pasupuleti, 2025): On board Edge AI drastically improves latency at the cost of excess computation to improve collision avoidance and route optimization. Implementing RL on locally embedded hardware allows the agent to not solely rely on the cloud and compute solutions much faster.

- (Anon., 2025): The multi-agent platform routing demonstrates that using multiple agents and clustering regions significantly boosts the model but drains the computational resources and battery life of the agents. The proposed idea of a mobile truck to serve as a restock station enables for a sustainable and flexible large-scale deployment of the drones enabling features like en route charging and flexible docking.

- (Jazairy, 2025): The heuristic findings suggest that drones significantly help in reduction of emission and leaves behind a miniscule carbon footprint as compared to a traditional delivery system. The review highlights their immense possibilities along with their limitation caused due to battery life, payload and safety regulation.

After weighing these approaches, (Qin, 2025) stands out as the most appealing and feasible model. Their solution explicitly addresses the energy constraints by clustering and dividing the objective among multiple drones. Evidence of it can seen in simulated environments where they outperform single-agent models in both efficiency and energy conservation. Managing computing resources among all the drones is already challenging enough so adding an additional agent as a docking station like (Anon., 2025) suggested just for an extended battery life might prove to be a behemoth of a task. However, (Pasupuleti, 2025) embedded RL coupled with (Qin, 2025) would drastically improve the latency and most likely outperform every other rigid models. Together, they provide strong evidence that on-board Edge AI with an orchestrated fleet of agents can most effectively meet the demands of autonomous complex systems.

5. CONCLUSION

This report illustrates that AI-driven pathfinding is vital for effective autonomous drones delivery. While optimization methods like temporal logic and multi-objective GA guarantees completion within the time constraints with tradeoffs, learning based dynamic approaches like muti-agent RL, embedded edge AI offer much more robustness and adaptability to real-world conditions. The final verdict remains that multi-agent models offer much more flexibility to better deal with real-world scenarios but a hybrid approach of with embedded edge AI framework could lead to a holistic model with improved latency and performance required for an ideal last-mile autonomous delivery network.

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